

Improving Medical Image Segmentation and Classification Using a Novel Joint Deep Learning Model

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Abstract

Although deep learning models have been widely used in medical imaging research field to perform lesion segmentation and classification tasks, several challenges remain to optimally apply deep learning models and improve model performance. The objective of this study is to investigate a new novel joint model and assess model performance improvement as increase of training dataset size. Specifically, we select and modify a novel J-Net as a joint model, which includes a two-way CNN architecture that combines a U-net model with an image classification model. A skin cancer dataset with 1200 images along with the annotated lesion masks and ground truth of “mild” and “severe” status is used. From this dataset, 11 subsets are randomly generated from 200 to 1200 images with an incremental rate of 100. Each subset is then divided into training, validation and testing groups using a ratio of 70:20:10, respectively. The performance of the new joint model is compared with two independent models to separately perform lesion segmentation and classification. The study results show when training the models using data subsets of 200 to 1200 images, accuracy levels increase from 0.80 to 0.92, or 0.86 to 0.95 in lesion segmentation. The lesion classification increases from 0.80 to 0.90, or 0.82 to 0.93 using two single models and one joint J-Net model, respectively. Thus, this study demonstrates that applying this new J-Net joint model enables to achieve higher lesion segmentation and classification accuracy than two single models. Additionally, model performance also increases as increase of training dataset size.

Keywords: Deep Learning, Semantic Segmentation, Image classification, Joint models, Hybrid learning, Skin Cancer.

1. INTRODUCTION

Applying computer-aided detection or diagnosis (CAD) schemes of medical images has attracted broad research interest in medical imaging informatics field [1-3]. The primary purpose of these CAD schemes is to assist clinicians and researchers in a range of applications, from organ segmentation [4-8] to disease diagnosis [9-12]. Most CAD schemes are developed using the conventional machine learning methods or models during the initial research stages, which often require domain expertise for image feature selection to build better CAD schemes. However, due to the difference or gap between human vision and computer vision, effectively or optimally defining or selecting non-redundant and highly clinically relevant handcrafted medical image features is difficult and thus remains a major challenged issue in developing CAD schemes.

However, recent studies have demonstrated that applying deep learning methods can successfully navigate or overcome the shortcomings mentioned above and thus it has been attracted extensive research in the past decade [13-15]. Deep Learning has enabled advancements in many fields, including computer vision [16, 17]. Computer vision is a collaborative scientific field that deals with how computers can be trained to gain a high-level understanding of digital images or videos. In recent years, deep learning models have been developed on a wide range of medical imaging to diagnose many diseases [28-34]. In addition, researchers have also tried to fuse or combine multiple deep learning models in order to achieve higher efficiency in image processing [25, 26]. These hybrid or joint models can process different data types and perform better than individually trained models. Collaborative models learn different versions of features from the data and boost performance. Thus, the insights processed and extracted from joint models aid in solving complex

problems. Especially, in the field of biomedical imaging, analyzing the image features from different models could help medical professionals to create a clear intuition of the issue at hand.

Despite the promising advantages of developing and applying deep learning models, the limited access to data in the medical domains hinders the application of deep learning models. Due to data regulations and market fragmentation, it can be challenging to fabricate large homogeneous datasets. Therefore, creating deep learning models that can work with datasets of limited size is still challenging. Alternatively, gathering additional information from a single image is another approach. For instance, in a traditional binary classification task, each image is designated a single 0/1 label. In supervised tasks, this label and input image are the only information given to a model to optimize its internal parameters. Thus, in small datasets, a lack of knowledge can lead to overfitting.

Providing additional information to the model by producing a segmentation mask of the input image can help to alleviate this overfitting problem. This mask typically has the same dimensionality as the input image and provides more information for the model to optimize on. Hence, it is possible to improve model performance on smaller datasets by combining classification and segmentation tasks as a single joint network. Y-Net [26] is one such model that takes full advantage of this approach to joint training. However, the network has not gained widespread appeal, partially because its effectiveness has yet to be demonstrated in independent, controlled studies. The objective of this study is to introduce a hybrid deep-learning model that simultaneously generates segmentation masks and diagnoses classification labels for skin cancer images. Our network provides a simple architecture that can be easy to adapt to any dataset. Due to the straightforward nature of our network, smaller datasets with a few hundreds of data records can yield highly accurate results avoiding overfitting.

2. MATERIALS & METHODS

2.1. Image Dataset

This experiment is tested on a dataset obtained from Kaggle, an online open-source platform for data. The dataset has skin cancer images with corresponding segmentation masks and a binary diagnostic label for the pigmented lesions on the skin. The segmentation masks are gray-scale images with white pixels (255) identified as the lesions on the skin and black pixels (0) as the normal skin. The binary label states the presence of melanocytic nevi (NV) in the skin images. This data field has 0 representing 'mild' and 1 representing the 'severe' presence of nevus cells in the corresponding image of skin cancer lesion. In total, 1200 images are selected with equal distribution of both the cases for the study.

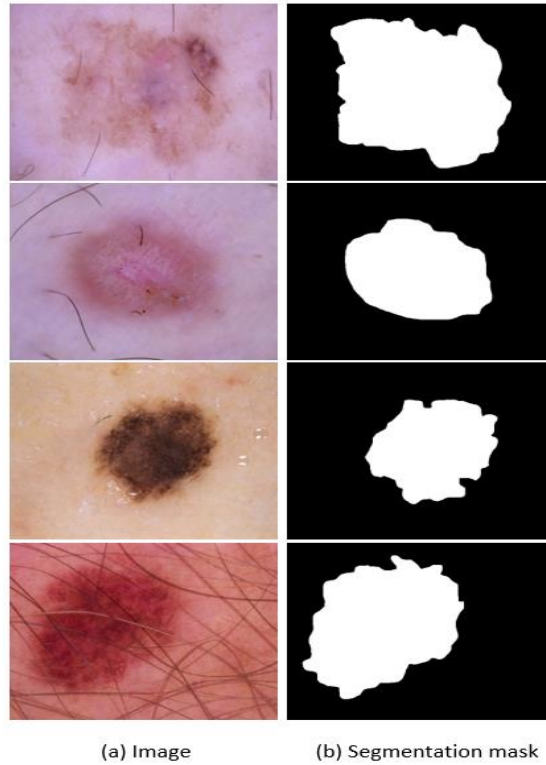


Figure 1: Skin cancer images and corresponding lesion masks

2.2. Methods

For the training of both U-Net and Y-Net models, the dataset is split into train, validation, and test ratio of 70:20:10. The images and masks are normalized for data standardization and converted to the size of 256X256. Both models are trained for 150 epochs with a learning rate of $3e-5$ and batch size 10 with an SGD optimizer. For the segmentation task, a custom loss function is developed using a Sigmoid function followed by a Binary cross entropy (BCE) loss function. To optimize the binary classification task, a BCE loss function is used.

2.2.1. U-Net

The U-Net architecture was developed for biomedical image segmentation that contains two phases [35]. The first phase is the contraction phase, also known as the encoder, which captures the context of the image. The encoder is a traditional stack of convolutional and max-pooling layers. The second phase is the symmetric expanding phase, also known as the decoder, which enables precise localization using transposed convolutions. Thus, it is an end-to-end, fully convolutional network (FCN).

For this study, the traditional U-Net architecture is modified by reducing the convolution block size to 32×32 on all the stages of the architecture. This modification reduced the complexity of the model and thus helped in handling overfitting. Each stage of the encoder has two convolution blocks, where each block has a 3×3 convolution layer followed by a batch normalization and a ReLU activation layer. A 2×2 max pooling layer is used to reduce the dimensionality of the features and passed to the next stage of the encoder. Four encoder stages reduce the image features from 256×256 to 16×16 , followed by a bottleneck layer where the features are up sampled and send into the decoder. The decoder blocks upscale the features from the bottom stage and pair with the corresponding encoder features. Following the decoder, a sigmoid layer is used to produce a binary segmentation mask.

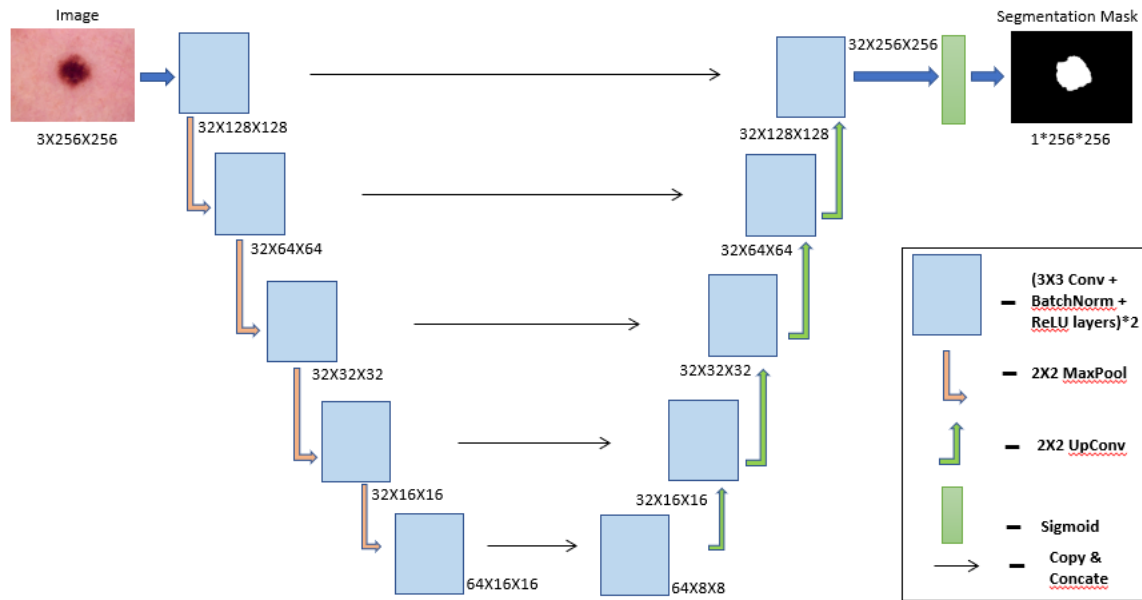


Figure 2: U-Net architecture

2.2.2. Binary Image Classifier

The second independent model used to perform image classification is a simple binary classifier. The classifier's architecture is developed to resemble the encoder part of the U-Net plus an additional convolution block followed by an average pool layer and linear layers producing a binary output. The output '1' meaning to severe and '0' meaning to the mild presence of melanoma on the lesions. For the model training, 1200 image dataset with a 70:20:10 data split for validation and test sets is drawn and the performance metrics are measured to compare with the results from the J-Net.

2.2.3. J-Net

A novel model, J-Net, is developed by combining the U-Net along with the binary image classifier, inspired by Y-Net [36]. The J-Net architecture contains two-way feature learning modules. The first module is a U-Net, a deep down sampling-to-up sampling sub-network for semantic features. The second is a convolutional sub-network without down sampling for classification features. The first module has the exact architecture of the U-net described in the previous section. The classification branch attached to the encoder part of U-Net uses the same convolution blocks as the encoder stages of the segmentation branch. An average pooling, three linear, and sigmoid layers are connected to the classification convolution blocks. Thus, the feature maps from the encoder phase of J-Net serves as the input to the binary classifier the classification branch categorizes the image into two classes.

The J-Net is optimized using the combined loss from segmentation ($Loss_{Seg}$) and classification tasks ($Loss_{Clas}$). Combining losses helps the network understand the features of each data label along with its segmentation mask. The additional information learned by the classification branch helps the segmentation branch in optimizing the masks and vice versa.

$$Loss_{JNet} = Loss_{Seg} + Loss_{Clas} \quad (1)$$

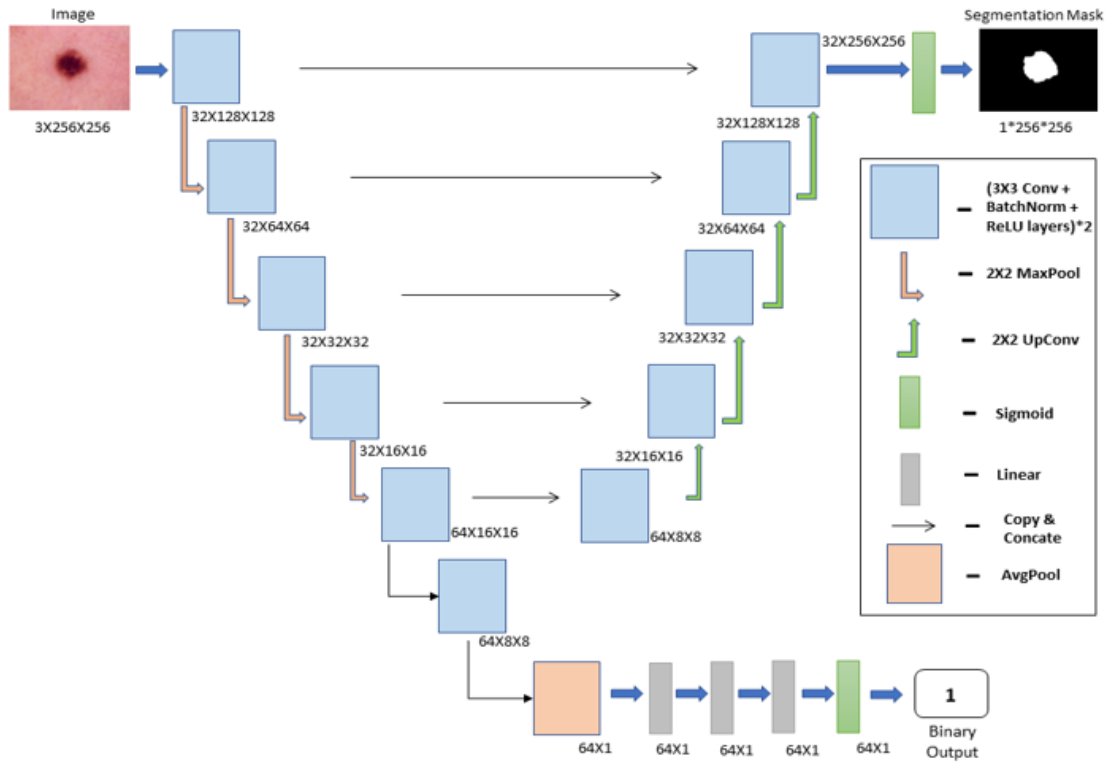


Figure 3: J-Net architecture

3. RESULTS

The results of the U-Net and the J-Net models are compared based on lesion segmentation and classification performance metrics. For the testing purpose, 10 percent of the dataset is used. From the observed results, J-Net has higher segmentation accuracy than U-Net over smaller datasets. Especially, J-Net accurately segments the contours of the images. Both models' segmentation metrics improved as the dataset's size increased from 200 to 1200 images.

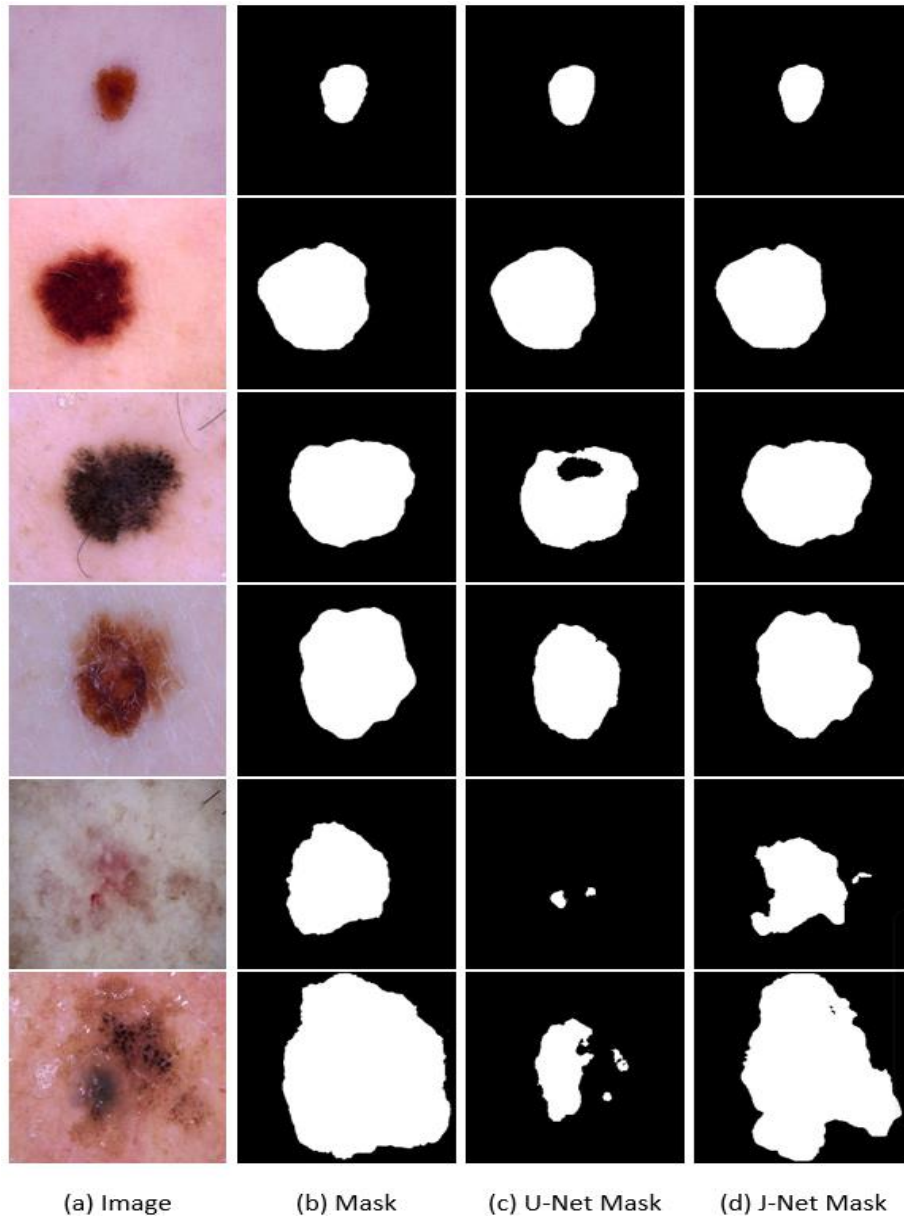


Figure 4: From left to right: skin cancer images, corresponding original segmentation masks, U-Net generated masks, and J-Net generated masks

To compute the statistical significance, the models are trained 30 times and the test set BCE loss is observed. The mean distribution of BCE loss for J-Net is lower than the U-Net in the segmentation task showing over 8% improvement.

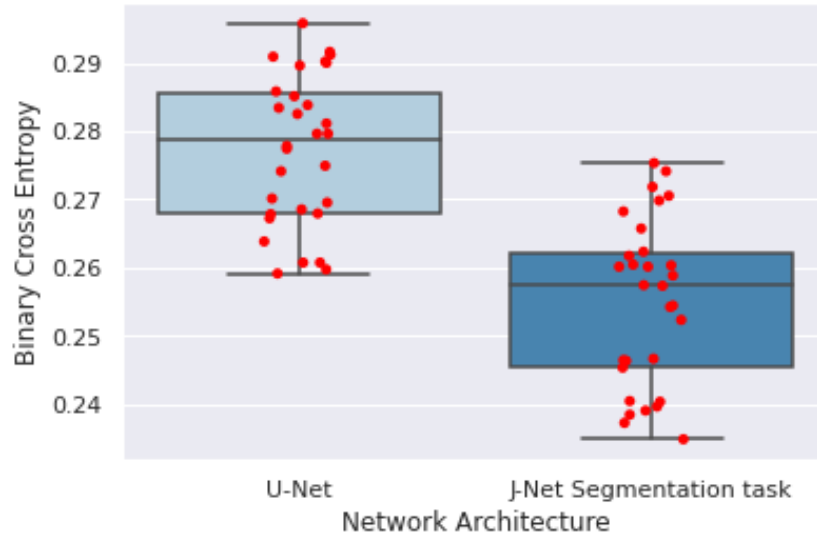


Figure 5: Boxplot results for 30 samples of BCE loss from lesion segmentation branch from U-Net vs J-Net classifier with $p < 0.05$

Similarly, the J-Net classifier outperformed a simple binary classifier with the same architecture with over 5% improvement on the BCE loss. This experiment shows that adding additional information in training models can help them understand the hidden patterns from all perspectives.

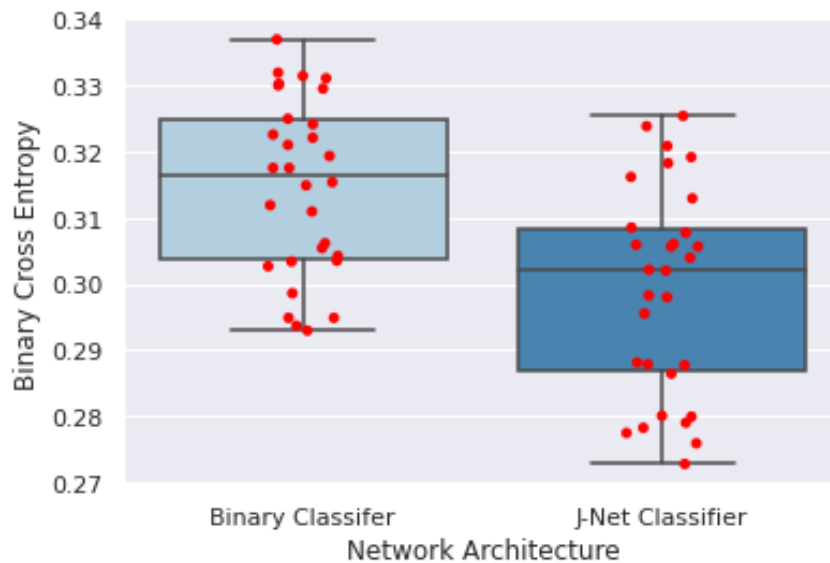


Figure 6: Boxplot results for 30 samples of BCE loss from binary image classifier vs J-Net classifier with $p < 0.05$

Both the models are evaluated with performance metrics for the lesion segmentation task. Performance metrics like dice score, intersection over union (IOU), accuracy, precision, recall, F1, and binary cross entropy loss scores are recorded with varying training datasets between the models.

Table 1: Segmentation task metrics results between U-Net and J-Net with varying training data size from 200 to 1200

size	U-Net Segmentation results				J-Net Segmentation results			
	Precision	Accuracy	Dice	IOU	Precision	Accuracy	Dice	IOU
1200	0.82	0.9206	0.9254	0.7608	0.8719	0.9589	0.9637	0.8078
1100	0.81	0.9113	0.9175	0.7534	0.8677	0.9376	0.9586	0.8022
1000	0.80	0.9078	0.9107	0.7485	0.8446	0.9310	0.9478	0.7969
900	0.7981	0.8914	0.9013	0.7471	0.8381	0.9274	0.9279	0.7841
800	0.7959	0.8676	0.8971	0.7366	0.8282	0.9113	0.9147	0.7713
700	0.7801	0.8459	0.8864	0.7357	0.8196	0.8998	0.9063	0.7671
600	0.7677	0.8305	0.8849	0.7301	0.8082	0.8895	0.8872	0.7505
500	0.7612	0.8282	0.8705	0.7294	0.8031	0.8783	0.8838	0.7443
400	0.7582	0.8207	0.8699	0.7273	0.7996	0.8740	0.8757	0.7386
300	0.7503	0.8056	0.8623	0.7238	0.7925	0.8661	0.8685	0.7321
200	0.7497	0.7993	0.8571	0.7206	0.7844	0.8604	0.8602	0.7284

The classification results between J-Net and a binary classification model are drawn for comparison. The binary classifier has the same architecture as the J-Net classifier to ensure the fairness of the results. The outcomes of the Y-Net classifier outperformed the binary classifier with better precision and Recall. Thus, improved in categorizing the images into mild and severe melanocytic nevi classes. This study can be easily extended to other domains with limited image data but with additional attributes about the images.

Table 2: Classification model performance metrics between Binary classifier and J-Net classifier with varying training data size from 200 to 1200

size	Binary Image Classifier results				J-Net Classifier results			
	Precision	Accuracy	Recall	F1	Precision	Accuracy	Recall	F1
1200	0.74	0.90	0.73	0.72	0.78	0.93	0.76	0.76
1100	0.74	0.89	0.72	0.72	0.77	0.92	0.76	0.76
1000	0.73	0.89	0.72	0.72	0.76	0.91	0.76	0.76
900	0.72	0.88	0.70	0.70	0.76	0.90	0.74	0.74
800	0.70	0.86	0.68	0.68	0.74	0.89	0.72	0.72
700	0.68	0.85	0.66	0.66	0.72	0.88	0.70	0.70
600	0.69	0.85	0.68	0.68	0.72	0.87	0.70	0.70
500	0.68	0.84	0.68	0.68	0.71	0.85	0.68	0.68
400	0.68	0.82	0.66	0.66	0.70	0.84	0.68	0.68
300	0.68	0.81	0.66	0.7238	0.69	0.82	0.68	0.67
200	0.66	0.80	0.65	0.7206	0.68	0.82	0.66	0.66

4. DISCUSSION

In this paper, we present a unique or novel study that analyzes and compares two methods of applying two independent models separately and one joint model to conduct lesion segmentation and classification tasks, as well as lesion segmentation and classification performance changes of these two methods. From the study results, we observe and discover the following two interesting aspects:

1. The J-Net joint model yields the higher lesion segmentation and classification accuracy than using two independent single models even with small datasets (as shown in Tables 1 and 2). This indicates that different deep learning models may have different focus in learning and extracting image features, thus the generated segmentation results and/or classification scores are not highly collected. As a result, this can also be explained due to the features learnt from two integrated branches of the J-Net model, which enables to help improve model performance or accuracy in both lesion segmentation and classification. Therefore, combining two tasks of lesion segmentation and classification into one model allows for better understanding and integrating image features and the context learned from different perspectives.
2. This study also demonstrates a clear trend of performance increase of the deep learning models including U-Net and joint J-Net models as the increase of training and testing dataset size (i.e., from 200 to 1200 in this study). Observation of such performance increasing trend clearly indicates that increasing training dataset size and diversity plays a very important role in developing higher accurate and more robust deep learning models using medical images. Thus, in future research, more effort should be added to increase training dataset size by either collecting more clinical images or developing more effective algorithms or models to produce more clinically relevant synthetic images.

With several limitations including using only 2D images of skin cancer and difficulty to obtain accurate lesion segmentation masks, we recognize that this is a quite unique study to address two important issues in optimally applying deep learning models in medical image research. Thus, distinctive studies are needed to investigate more robust and multi-usage of data for different medical imaging applications in the future.

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